



SCHOOL *of* DATA SCIENCE



Containers
Virtualization and the linux kernel



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Containers

Leveraging the Linux kernel

- Virtualization
- What is a container?
- Container versus Virtual Machine

■ Container Definition

A standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably from one computing environment to another

■ Comparison to Virtual Machine

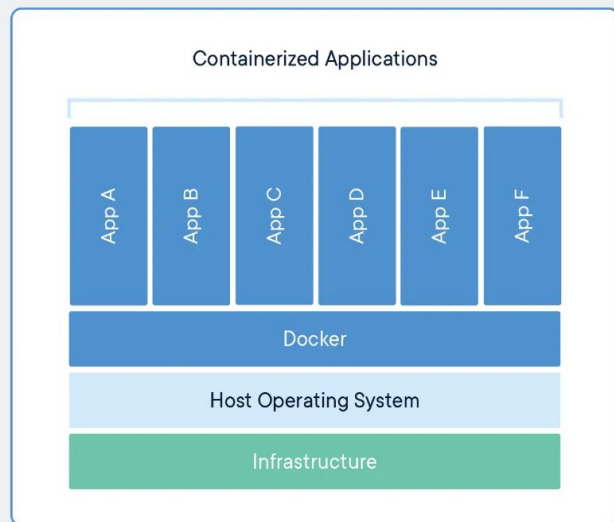
Virtualization is the running of a machine inside a machine

- Unlike a 'slice', a VM carries with it an operating system.
- The OS is run like an application, as far as the host, it's just like any other app.
- As with the slice approach we can share disk, open ports etc.

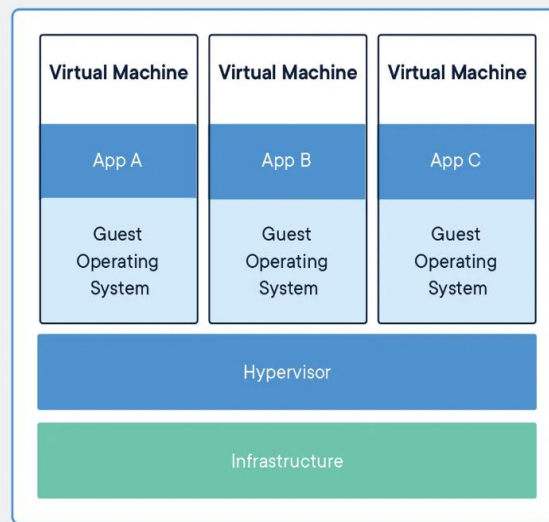
Container vs Virtual Machine

Container's jail is like having a "curtain" between processes.

Think Horizontal



CONTAINERS



VIRTUAL MACHINES

Virtual machines as stacking on TOP of hypervisor.

Think Vertically

[source](#)

Containers are like users (almost)

A process called 'jailing' keeps processes from running in same space

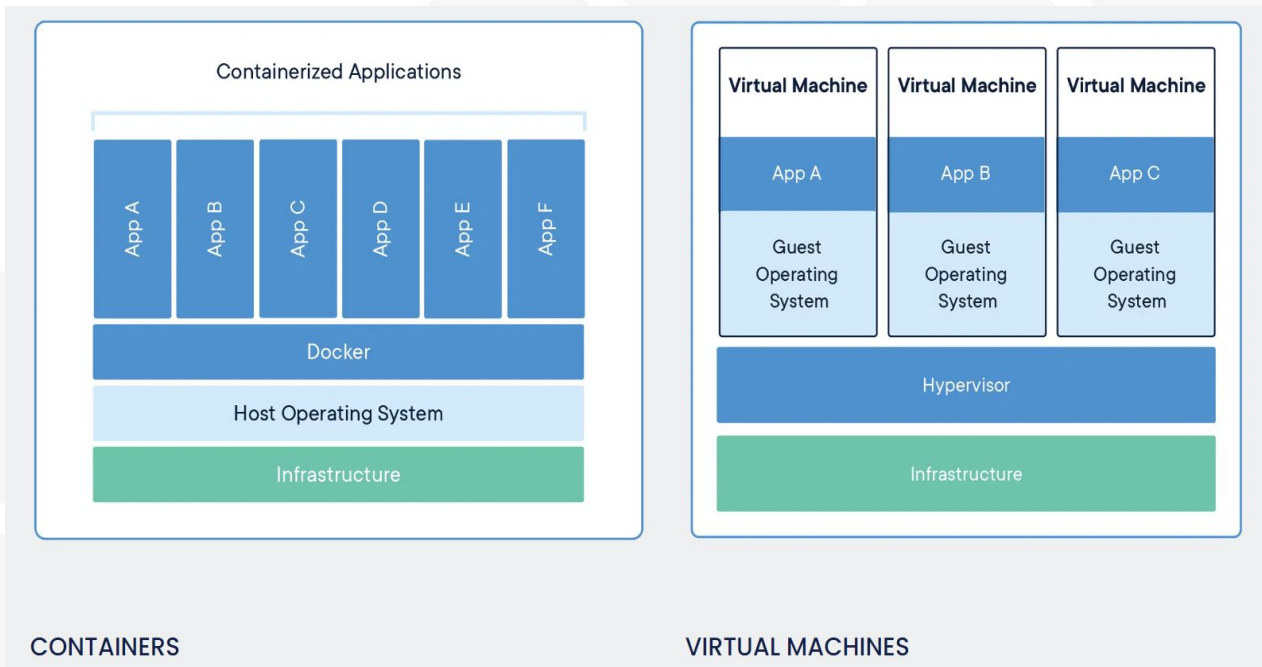
- A 'jailed' space is created re-using the linux kernel
- Linux versions are a layer on top of a common kernel
- We can configure the slice to interact with disk space, and => **ports**.

Container vs Virtual Machine

Apps all start from same kernel.

Thus are linux like.

Can run 'windows' apps, but at bottom are linux. (New windows containers) [source](#)



Virtual Machines are a fully running app simulating an OS. The Hypervisor is like an “OS for OSs”

VMs can be ANY type of OS

■ Pros and cons

Container pros

- Lightweight, since they only carry the 'extra' code
- Turn on and off really fast
- Really secure, 'jailing' was intended to make processes isolated

By comparison

- VMs are usually large files, carrying the full OS
- Slower to boot
- Slower to run

■ Pros and cons II

Container cons

- Since they rely on kernel, they are all **'linux like'**
- Usually rely on command line, since slice would conflict with host UI

By comparison

- VMs can run any OS that has been adapted, you can run windows on linux, or Mac on windows etc.
- Each VM can have it's own UI. The experience is complete.

What is Amazon's EC2?

Hypervisor = An application specifically to create and run VMS

Server = A larger than usual computer, with lots of disk and RAM, designed specifically to run vms or containers

Data Center = A building full of Servers running Hypervisors

When you 'rent' an EC2 machine you are basically timesharing a VM on a hypervisor.

■ But why do we want containers anyway?

- Reproducibility: Make the “it runs on my system” problem go away.
- Deploy “ready to run” applications, since the container can carry dependencies and configuration
- Reproducing development environments, and testing environments.
- Easy to “**compose**” a system of machines, a multi app system.

An example of a composed system is the 3-tier Front end load balancer, with app in the middle and database back-end.

■ Docker, the primary container application

- Docker is an app you can run at the command line to download and run containers
- Docker has VMs running on Mac and Windows in order to run the containers
- Dockerhub - hosts containers so we can version and share

[DOCKERHUB SEARCH](#)

Closeup on linux 'ports'

Computers exchange messages with each other across numbered 'ports'.

A computer can open or close ports for security.

By agreement, only certain types of messages are sent/received on specific ports. For example

- Http, port **80**
- MySQL, port **3306**

We CONNECT a container by opening ports

Cloud Computing has exploded with Docker and VMs

- Easy to open an account and use EC2. You can set up
 - Web servers
 - Jenkins
 - Databases
 - pyspark processors
- Watch the costs on EC2 (VMS)
 - AWS charges you by the hour/minute depending on size.
 - A Running machine is the most expensive
 - A machine with detachable disk can be stopped. You then pay for 'storage only' in between sessions. When you 'terminate' it, your storage also goes away.
- Docker backs services like AWS Lambda.
 - Your code/script is stored in AWS.
 - When you 'trigger' it, a docker container is set up and runs your code.
 - Pay less than a cent for execution for small scripts.

This weeks LAB

In our lab this week:

- We will log into our AWS accounts
- Create a virtual machine for our use during course
- Update the virtual machine to present package status
- Load packages we need
- Load github keys, and set up github